

Virtual University of Pakistan

Regional Campus PLHR-49 Allama Iqbal Road, Lahore

CS101 MIDTERM 2026 MCQS FILES

Topic: Introduction to Computer Science

Q1. What is the physical part of a computer system called?

- A. Software
- B. Hardware
- C. Database
- D. Algorithms

Correct Option: B (Hardware)

Topic: Search Engines

Q2. Which symbol is used to search on a specific social media platform in Google?

- A. #
- B. &
- C. @
- D. %

Correct Option: C (@)

Topic: Roots of Computing

Q3. Which early computing device was created in ancient China and uses beads on rods?

- A. ENIAC
- B. Abacus
- C. Pascal's Calculator
- D. Punch Cards

Correct Option: B (Abacus)

Topic: Roots of Computing

Q4. Which machine developed in the 1940s used 20,000 vacuum tubes?

- A. ENIAC
- B. Abacus
- C. Mark I
- D. Gears

Correct Option: A (ENIAC)

Topic: Bits

Q5. How many bits make up one byte?

- A. 4
- B. 8
- C. 16
- D. 32

Correct Option: B (8)

Topic: Boolean Operations

Q6. Which Boolean operation outputs 1 only when both inputs are 1?

- A. OR
- B. XOR
- C. AND
- D. NOT

Correct Option: C (AND)

Topic: Boolean Operations

Q7. Which Boolean operation outputs 1 when inputs are different?

- A. AND
- B. NOT
- C. OR
- D. XOR

Correct Option: D (XOR)

Topic: Hexadecimal Notation

Q8. In hexadecimal notation, a single symbol represents how many bits?

- A. 2
- B. 4
- C. 8
- D. 16

Correct Option: B (4)

Topic: Hexadecimal Notation

Q9. What is the hexadecimal representation of the binary pattern 1010?

- A. 8
- B. 9
- C. A
- D. B

Correct Option: C (A)

Topic: Storing a Bit

Q10. What type of main memory allows cells to be accessed independently in any order?

- A. Magnetic Tape
- B. ROM
- C. RAM
- D. CD

Correct Option: C (RAM)

Topic: Magnetic Systems

Q11. The time required to move the read/write heads from one track to another on a magnetic disk is called:

- A. Rotation Delay
- B. Seek Time
- C. Access Time
- D. Transfer Rate

Correct Option: B (Seek Time)

Topic: Magnetic Systems

Q12. The sum of seek time and rotation delay is known as:

- A. Transfer Rate
- B. Latency
- C. Access Time
- D. Execution Time

Correct Option: C (Access Time)

Topic: Optical System

Q13. Which of the following uses a blue-violet laser to store up to five times more data than DVDs?

- A. CD
- B. Magnetic Disk
- C. Blu-ray Disks (BDs)
- D. Flash Drive

Correct Option: C (Blu-ray Disks (BDs))

Topic: Flash Drives

Q14. How are bits stored in flash drive technology?

- A. By burning holes in a disk
- B. By aligning magnetic fields
- C. By trapping electrons in silicon dioxide chambers
- D. By spinning optical plates

Correct Option: C (By trapping electrons in silicon dioxide chambers)

Topic: Representing Text

Q15. How many bits are traditionally used for a standard ASCII character?

- A. 7
- B. 8
- C. 16
- D. 32

Correct Option: A (7)

Topic: Representing Text

Q16. Which standard uses 21 bits to support characters of multiple international languages?

- A. ASCII
- B. Unicode
- C. UTF-8
- D. EBCDIC

Correct Option: B (Unicode)

Topic: Representing Images

Q17. What is the short form for 'Picture Element'?

- A. Dot
- B. Pix
- C. Pixel
- D. Point

Correct Option: C (Pixel)

Topic: Representing Sound

Q18. Which of the following encodes directions for producing music rather than storing the sound itself?

- A. MP3
- B. MIDI
- C. WAV
- D. FLAC

Correct Option: B (MIDI)

Topic: Binary Notation

Q19. What is the decimal equivalent of the binary number 1011?

- A. 9
- B. 10
- C. 11
- D. 12

Correct Option: C (11)

Topic: 2's Complement Notation

Q20. In two's complement notation, the leftmost bit is primarily used as the:

- A. Sign bit
- B. Exponent bit
- C. Mantissa
- D. Parity bit

Correct Option: A (Sign bit)

Topic: Floating Point Notation

Q21. In floating point notation, the field that determines the position of the radix point is called:

- A. Mantissa
- B. Sign bit
- C. Exponent
- D. Radix

Correct Option: C (Exponent)

Topic: Truncation Errors

Q22. Which error occurs when a value cannot be stored accurately because the mantissa field is not large enough?

- A. Syntax error
- B. Logic error
- C. Truncation error
- D. Overflow error

Correct Option: C (Truncation error)

Topic: Data Compression

Q23. Which data compression technique replaces sequences of identical data elements with a code indicating the element and its count?

- A. Huffman encoding
- B. Dictionary encoding
- C. Run-length encoding
- D. Relative encoding

Correct Option: C (Run-length encoding)

Topic: Data Compression

Q24. GIF is an example of which type of image compression system?

- A. Dictionary encoding
- B. Run-length encoding
- C. Differential encoding
- D. LZW encoding

Correct Option: A (Dictionary encoding)

Topic: CPU Basic

Q25. Which part of the CPU contains the circuitry that performs addition and subtraction?

- A. Control Unit
- B. Registers
- C. Arithmetic/Logic Unit
- D. Bus

Correct Option: C (Arithmetic/Logic Unit)

Topic: CPU Basic

Q26. The collection of wires connecting the CPU and main memory is called:

- A. Bus
- B. Link
- C. Cable
- D. Port

Correct Option: A (Bus)

Topic: Stored Program

Q27. The concept of storing a computer's program in its main memory is called:

- A. Stored-program concept
- B. Pipelining
- C. Caching
- D. DMA

Correct Option: A (Stored-program concept)

Topic: CPU Architecture Philosophies

Q28. A CPU designed to execute a minimal set of machine instructions is known as:

- A. CISC
- B. RISC
- C. ARM
- D. Intel

Correct Option: B (RISC)

Topic: Machine Instruction Categories

Q29. An instruction that copies data from a memory cell into a CPU register is known as a:

- A. STORE instruction
- B. LOAD instruction
- C. ADD instruction
- D. JUMP instruction

Correct Option: B (LOAD instruction)

Topic: Program Execution

Q30. Which special-purpose register contains the address of the next instruction to be executed?

- A. Instruction Register
- B. Program Counter
- C. Accumulator
- D. Memory Address Register

Correct Option: B (Program Counter)

Topic: Program Execution

Q31. What are the three steps of the machine cycle?

- A. Read, Write, Execute
- B. Fetch, Decode, Execute
- C. Input, Process, Output
- D. Load, Add, Store

Correct Option: B (Fetch, Decode, Execute)

Topic: Logic Operators

Q32. Which bitwise operation is commonly used for masking to place 0s in one part of a bit pattern?

- A. OR
- B. XOR
- C. NOT
- D. AND

Correct Option: D (AND)

Topic: Rotation and Shift

Q33. Shifting binary digits to the left corresponds to which mathematical operation?

- A. Addition
- B. Subtraction
- C. Multiplication by two
- D. Division by two

Correct Option: C (Multiplication by two)

Topic: Role of Controller

Q34. Which intermediary apparatus handles communication between a computer and other external devices?

- A. CPU
- B. Controller
- C. Bus
- D. Memory

Correct Option: B (Controller)

Topic: Direct Memory Access

Q35. The ability of a controller to access main memory while the CPU is not using the bus is known as:

- A. Interrupt
- B. Pipelining
- C. DMA
- D. Multitasking

Correct Option: C (DMA)

Topic: Communication media

Q36. In which type of communication are several signals transferred at the same time, each on a separate line?

- A. Serial communication
- B. Parallel communication
- C. Wireless communication
- D. Fiber optic communication

Correct Option: B (Parallel communication)

Topic: Pipelining

Q37. The technique of allowing steps in the machine cycle to overlap to increase throughput is called:

- A. Pipelining
- B. Multiprogramming
- C. Time-sharing
- D. Batch processing

Correct Option: A (Pipelining)

Topic: Operating Systems: History

Q38. The execution of jobs by collecting them in a single group and executing them without user interaction is called:

- A. Time-sharing
- B. Interactive processing
- C. Batch processing
- D. Real-time processing

Correct Option: C (Batch processing)

Topic: Operating Systems: Basic Concepts

Q39. The procedure of changing the CPU's attention from one process to another is called:

- A. Time-slicing
- B. Booting
- C. Process switch
- D. Interrupt handling

Correct Option: C (Process switch)

Topic: Operating Systems: Deadlock

Q40. A condition in which two or more processes are blocked from progressing because each is waiting for a resource allocated to another is called:

- A. Starvation
- B. Deadlock
- C. Interrupt
- D. Multitasking

Correct Option: B (Deadlock)

Topic: Networking and the Internet

Q41. A network of computers connected for the purpose of sharing resources is called a:

- A. Bus
- B. Computer network
- C. Controller
- D. CPU

Correct Option: B (Computer network)

Topic: Algorithm: Formal Definition

Q42. An ordered set of unambiguous, executable steps that defines a terminating process is called an:

- A. Algorithm
- B. Program
- C. Process
- D. Instruction

Correct Option: A (Algorithm)

Topic: Algorithm: Representation

Q43. A well-defined building block from which algorithm representations can be constructed is called a:

- A. Syntax
- B. Primitive
- C. Semantic
- D. Code

Correct Option: B (Primitive)

Topic: Algorithm: Pseudocode

Q44. A less formal, intuitive notational system used to express algorithms during development is called:

- A. Machine language
- B. High-level language
- C. Pseudocode
- D. Binary code

Correct Option: C (Pseudocode)

Topic: Algorithm: Discovery

Q45. The top-down methodology of breaking a problem into smaller, easily solvable subproblems is called:

- A. Stepwise refinement
- B. Recursion
- C. Iteration
- D. Bottom-up design

Correct Option: A (Stepwise refinement)

Topic: Iterative Structures

Q46. The three activities required for successful loop control are initialize, modify, and:

- A. Execute
- B. Test
- C. Halt
- D. Fetch

Correct Option: B (Test)

Topic: Iterative Structures

Q47. A loop where the test for termination is performed before the body is executed is called a:

- A. Posttest loop
- B. Infinite loop
- C. Pretest loop
- D. Recursive loop

Correct Option: C (Pretest loop)

Topic: Insertion Sort Algorithm

Q48. Which sorting algorithm sorts a list by shuffling its entries analogous to sorting index cards on a desk?

- A. Sequential Sort
- B. Binary Sort
- C. Insertion Sort
- D. Selection Sort

Correct Option: C (Insertion Sort)

Topic: Recursive Structure

Q49. The technique where a repetitive set of instructions is executed as a subtask of itself is known as:

- A. Looping
- B. Recursion
- C. Iteration
- D. Branching

Correct Option: B (Recursion)

Topic: Algorithm Efficiency

Q50. Which search algorithm interrogates at most $\log_2(n)$ entries in its worst-case scenario?

- A. Sequential Search
- B. Linear Search
- C. Binary Search
- D. Insertion Search

Correct Option: C (Binary Search)

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Q51. Which of the following is considered the intangible, logical component of a computer system?

- A. Hardware
- B. Peripheral
- C. Software
- D. Motherboard

Correct Option: C (Software)

Topic: Introduction to Computer Science

Q52. What does the abbreviation 'PC' commonly stand for in computing?

- A. Public Computer
- B. Private Computer
- C. Personal Computer
- D. Portable Computer

Correct Option: C (Personal Computer)

Topic: Search Engines

Q53. Which operator is typically used in search engines to find an exact phrase?

- A. Asterisk (*)
- B. Quotation Marks ("")
- C. Plus (+)
- D. Tilde (~)

Correct Option: B (Quotation Marks (""))

Topic: Search Engines

Q54. Which symbol is commonly used to exclude a specific word from search engine results?

- A. Minus (-)
- B. Slash (/)
- C. Hash (#)
- D. Ampersand (&)

Correct Option: A (Minus (-))

Topic: Roots of Computing

Q55. Who is often widely recognized as the 'Father of Computing'?

- A. Alan Turing
- B. Charles Babbage
- C. John von Neumann
- D. Blaise Pascal

Correct Option: B (Charles Babbage)

Topic: Roots of Computing

Q56. Augusta Ada Byron is historically recognized as the world's first:

- A. Hardware Engineer
- B. System Analyst
- C. Programmer
- D. Network Administrator

Correct Option: C (Programmer)

Topic: Roots of Computing

Q57. Which mechanical computer was designed, but never fully completed, by Charles Babbage?

- A. ENIAC
- B. UNIVAC
- C. Analytical Engine
- D. Z3

Correct Option: C (Analytical Engine)

Topic: Boolean Operations

Q58. Which Boolean operation outputs 0 only when both of its inputs are 0?

- A. AND
- B. OR
- C. XOR
- D. NOT

Correct Option: B (OR)

Topic: Boolean Operations

Q59. What is the output of the Boolean NOT operator if the provided input is 1?

- A. -1
- B. 0
- C. 1
- D. Null

Correct Option: B (0)

Topic: Boolean Operations

Q60. Which logic gate produces a 0 output only when all of its inputs are 1?

- A. NAND
- B. NOR
- C. XOR
- D. OR

Correct Option: A (NAND)

Topic: Hexadecimal Notation

Q61. In the hexadecimal numbering system, the digit 'F' represents which decimal value?

- A. 10
- B. 12
- C. 14
- D. 15

Correct Option: D (15)

Topic: Hexadecimal Notation

Q62. How many bits in total are required to represent the hexadecimal value 'FF'?

- A. 4 bits
- B. 8 bits
- C. 16 bits
- D. 32 bits

Correct Option: B (8 bits)

Topic: Storing a Bit

Q63. An electronic circuit specifically designed to store a single bit of data is called a:

- A. Capacitor
- B. Resistor
- C. Flip-flop
- D. Transistor

Correct Option: C (Flip-flop)

Topic: Main Memory

Q64. Which type of memory is volatile, meaning it loses its contents when power is turned off?

- A. ROM
- B. Flash Memory
- C. Magnetic Disk
- D. RAM

Correct Option: D (RAM)

Topic: Main Memory

Q65. Which memory retains its programmed data permanently or semi-permanently even without power?

- A. RAM
- B. Cache
- C. ROM
- D. Registers

Correct Option: C (ROM)

Topic: Magnetic Systems

Q66. A magnetic disk's surface is logically divided into concentric circles called:

- A. Sectors
- B. Tracks
- C. Cylinders
- D. Blocks

Correct Option: B (Tracks)

Topic: Magnetic Systems

Q67. A single track on a magnetic disk is further divided into smaller, addressable segments known as:

- A. Sectors
- B. Clusters
- C. Platters
- D. Heads

Correct Option: A (Sectors)

Topic: Optical System

Q68. Optical discs like CDs and DVDs store data using a pattern of microscopic reflective areas called lands and:

- A. Holes
- B. Spots
- C. Pits
- D. Dots

Correct Option: C (Pits)

Topic: Optical System

Q69. What is the typical approximate storage capacity of a standard CD-ROM?

- A. 1.44 MB
- B. 700 MB
- C. 4.7 GB
- D. 25 GB

Correct Option: B (700 MB)

Topic: Flash Drives

Q70. Flash memory technology is fundamentally a specialized type of:

- A. SRAM
- B. DRAM
- C. EEPROM
- D. Magnetic Storage

Correct Option: C (EEPROM)

Topic: Representing Text

Q71. The EBCDIC character encoding scheme was primarily developed and utilized by which corporation?

- A. Microsoft
- B. Apple
- C. IBM
- D. Intel

Correct Option: C (IBM)

Topic: Representing Text

Q72. Unlike a simple text file, a word processor document contains text along with:

- A. Executable code
- B. Formatting information
- C. Video streams
- D. Operating system drivers

Correct Option: B (Formatting information)

Topic: Representing Images

Q73. Digital images represented as a grid or collection of individual pixels are formally known as:

- A. Vector graphics
- B. Bitmaps / Raster graphics
- C. 3D Models
- D. Fractals

Correct Option: B (Bitmaps / Raster graphics)

Topic: Representing Images

Q74. A graphic format that stores images mathematically as geometric shapes rather than pixels is called a:

- A. JPEG
- B. GIF
- C. Vector graphic
- D. Raster graphic

Correct Option: C (Vector graphic)

Topic: Representing Sound

Q75. The number of times the amplitude of a sound wave is captured and measured per second is called the:

- A. Bit rate
- B. Frequency
- C. Sampling rate
- D. Bandwidth

Correct Option: C (Sampling rate)

Topic: Representing Video

Q76. What psychological illusion creates the human perception of continuous motion in digital video?

- A. Persistence of vision
- B. Refraction
- C. Optical illusion
- D. Color blending

Correct Option: A (Persistence of vision)

Topic: Binary Addition

Q77. In basic binary addition, what is the exact mathematical result of adding 1 + 1?

- A. 2
- B. 10
- C. 11
- D. 0 with a borrow

Correct Option: B (10)

Topic: Errors in Arithmetic

Q78. What error condition occurs when a calculation results in a value too large to fit in the allotted memory space?

- A. Underflow
- B. Overflow
- C. Truncation
- D. Syntax Error

Correct Option: B (Overflow)

Topic: 2's Complement Notation

Q79. What is the 2's complement representation of the 4-bit binary number 0011?

- A. 1100
- B. 1101
- C. 1011
- D. 0100

Correct Option: B (1101)

Topic: 2's Complement Notation

Q80. In 2's complement, if the leftmost sign bit is '1', it indicates that the numerical value is:

- A. Positive
- B. Zero
- C. Negative
- D. Infinite

Correct Option: C (Negative)

Topic: Data Compression

Q81. Compression techniques that permanently discard some minor data to achieve much smaller file sizes are known as:

- A. Lossless compression
- B. Lossy compression
- C. Run-length encoding
- D. Huffman coding

Correct Option: B (Lossy compression)

Topic: Data Compression

Q82. Archive files like ZIP commonly utilize which broad category of compression?

- A. Lossy compression
- B. Lossless compression
- C. Video compression
- D. Audio compression

Correct Option: B (Lossless compression)

Topic: Data Compression

Q83. JPEG is a widespread compression standard primarily formulated and used for:

- A. Text documents
- B. Photographs/Images
- C. Databases
- D. Source code

Correct Option: B (Photographs/Images)

Topic: Data Compression

Q84. MPEG is an established ISO standard utilized mostly for the compression of:

- A. Text files
- B. Vector graphics
- C. Video and Audio
- D. Executables

Correct Option: C (Video and Audio)

Topic: CPU Basic

Q85. The high-speed memory locations inside a CPU that hold temporary data during calculations are called:

- A. Cache
- B. General-purpose registers
- C. Main Memory
- D. Virtual Memory

Correct Option: B (General-purpose registers)

Topic: CPU Basic

Q86. Which internal CPU component is responsible for decoding fetched instructions and managing execution?

- A. Arithmetic/Logic Unit
- B. General-purpose registers
- C. Control Unit
- D. System Bus

Correct Option: C (Control Unit)

Topic: Machine Instruction Categories

Q87. A machine instruction intended to copy data from a CPU register back into main memory is called:

- A. LOAD
- B. STORE
- C. JUMP
- D. ADD

Correct Option: B (STORE)

Topic: Machine Instruction Categories

Q88. An instruction that alters the normal sequential execution of a program by pointing to a new address is a:

- A. HALT
- B. MOVE
- C. JUMP
- D. AND

Correct Option: C (JUMP)

Topic: Peripheral Devices

Q89. Which of the following hardware devices operates strictly as an input device?

- A. Monitor
- B. Printer
- C. Keyboard
- D. Speaker

Correct Option: C (Keyboard)

Topic: Peripheral Devices

Q90. Which hardware device is specifically used to convert physical, printed documents into a digital format?

- A. Projector
- B. Scanner
- C. Plotter
- D. Modem

Correct Option: B (Scanner)

Topic: Handshaking

Q91. The preliminary two-way communication process establishing rules before devices transfer data is known as:

- A. Pipelining
- B. Broadcasting
- C. Handshaking
- D. Multiplexing

Correct Option: C (Handshaking)

Topic: Operating Systems: History

Q92. An operating system design that allows multiple users to interact with a central computer simultaneously is called:

- A. Batch processing
- B. Time-sharing
- C. Real-time processing
- D. Single-tasking

Correct Option: B (Time-sharing)

Topic: Software Classification

Q93. Software that explicitly manages computer hardware and provides a fundamental platform for other applications is an:

- A. Application Software
- B. Utility Software
- C. Operating System
- D. Firmware

Correct Option: C (Operating System)

Topic: System Software

Q94. Specialized programs that help perform maintenance, diagnostic, and organizational tasks on a computer are called:

- A. Drivers
- B. Utility software
- C. Word processors
- D. Spreadsheets

Correct Option: B (Utility software)

Topic: Operating Systems: Architecture

Q95. The absolute core component of an Operating System that manages basic system operations is called the:

- A. Shell
- B. User Interface
- C. Kernel
- D. Compiler

Correct Option: C (Kernel)

Topic: Memory Management

Q96. The OS technique of moving inactive program pages between RAM and secondary storage to free up space is called:

- A. Paging/Swapping
- B. Defragmentation
- C. Formatting
- D. Booting

Correct Option: A (Paging/Swapping)

Topic: Memory Management

Q97. Virtual memory creates an illusion of a much larger RAM by utilizing a dedicated portion of the:

- A. CPU Cache
- B. ROM
- C. Hard Disk / Mass Storage
- D. Optical Drive

Correct Option: C (Hard Disk / Mass Storage)

Topic: Algorithm: Representation

Q98. A standard graphical representation of an algorithm using interconnected specific shapes and symbols is called a:

- A. Pseudocode
- B. Flowchart
- C. Bar graph
- D. Pie chart

Correct Option: B (Flowchart)

Topic: Algorithm: Representation

Q99. In standard flowchart notation, a diamond-shaped box typically represents:

- A. Start/End
- B. Input/Output
- C. A decision / conditional operation
- D. A processing step

Correct Option: C (A decision / conditional operation)

Topic: Algorithm: Formal Definition

Q100. A defined, step-by-step mathematical or logical procedure designed to solve a specific problem is best defined as:

- A. An interface
- B. An operating system
- C. An algorithm
- D. A compiler

Correct Option: C (An algorithm)

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Topic: Search Engines

Q101. Which operator would you use to search for a specific file type, like a PDF?

- A. ext:
- B. filetype:
- C. type:
- D. format:

Correct Option: B (filetype:)

Topic: Search Engines

Q102. What is the primary function of a web crawler in a search engine?

- A. To index web pages
- B. To design web interfaces
- C. To block ads
- D. To compile code

Correct Option: A (To index web pages)

Topic: Roots of Computing

Q103. Which early device used punched cards to control the weaving of patterns in fabric?

- A. Jacquard Loom
- B. Analytical Engine
- C. Pascaline
- D. Turing Machine

Correct Option: A (Jacquard Loom)

Topic: Roots of Computing

Q104. The first generation of modern computers utilized which core technology?

- A. Transistors
- B. Integrated Circuits
- C. Vacuum Tubes
- D. Microprocessors

Correct Option: C (Vacuum Tubes)

Topic: Roots of Computing

Q105. Which generation of computers introduced the use of transistors, making them smaller and more reliable?

- A. First Generation
- B. Second Generation
- C. Third Generation
- D. Fourth Generation

Correct Option: B (Second Generation)

Topic: Bits

Q106. What is the highest decimal number that can be represented using 3 bits?

- A. 3
- B. 7
- C. 8
- D. 15

Correct Option: B (7)

Topic: Bits

Q107. How many distinct states can be represented by a 4-bit sequence?

- A. 8
- B. 12
- C. 16
- D. 32

Correct Option: C (16)

Topic: Boolean Operations

Q108. If input A is 1 and input B is 0, what is the output of A AND B?

- A. 0
- B. 1
- C. -1
- D. Null

Correct Option: A (0)

Topic: Boolean Operations

Q109. Which gate outputs a 1 if at least one of its inputs is 1?

- A. AND
- B. XOR
- C. NOT
- D. OR

Correct Option: D (OR)

Topic: Hexadecimal Notation

Q110. What is the binary equivalent of the hexadecimal digit 'B'?

- A. 1010
- B. 1011
- C. 1100
- D. 1101

Correct Option: B (1011)

Topic: Hexadecimal Notation

Q111. How do you represent the decimal number 16 in hexadecimal notation?

- A. F
- B. 10
- C. 16
- D. 1A

Correct Option: B (10)

Topic: Storing a Bit

Q112. Which circuit can maintain its state and is used as a fundamental building block of SRAM?

- A. Multiplexer
- B. Decoder
- C. Flip-flop
- D. Adder

Correct Option: C (Flip-flop)

Topic: Main Memory

Q113. Which part of the computer's memory is used to store the startup instructions (BIOS)?

- A. RAM
- B. Cache
- C. Virtual Memory
- D. ROM

Correct Option: D (ROM)

Topic: Main Memory

Q114. The process of continuously rewriting data in DRAM to prevent it from fading is called:

- A. Swapping
- B. Paging
- C. Refreshing
- D. Booting

Correct Option: C (Refreshing)

Topic: Magnetic Systems

Q115. In a magnetic disk drive, the mechanism that holds the read/write heads is called the:

- A. Spindle
- B. Access Arm
- C. Platter
- D. Sector

Correct Option: B (Access Arm)

Topic: Optical System

Q116. Which of the following optical formats generally has the highest data storage capacity?

- A. CD
- B. DVD
- C. Blu-ray
- D. Floppy Disk

Correct Option: C (Blu-ray)

Topic: Flash Drives

Q117. Flash memory is immune to physical shock primarily because it has:

- A. A metallic casing
- B. No moving parts
- C. Magnetic shielding
- D. Vacuum tubes

Correct Option: B (No moving parts)

Topic: Representing Text

Q118. ASCII originally utilized a 7-bit code. How many unique characters could it represent?

- A. 64
- B. 128
- C. 256
- D. 512

Correct Option: B (128)

Topic: Representing Text

Q119. Which modern encoding system was designed to unify the representation of characters from all human languages?

- A. ASCII
- B. EBCDIC
- C. Unicode
- D. ANSI

Correct Option: C (Unicode)

Topic: Representing Images

Q120. The number of bits used to represent the color of a single pixel is known as:

- A. Resolution
- B. Color depth
- C. Frame rate
- D. Aspect ratio

Correct Option: B (Color depth)

Topic: Representing Images

Q121. In the RGB color model, what color is created if Red, Green, and Blue are all set to their maximum values?

- A. Black
- B. White
- C. Gray
- D. Transparent

Correct Option: B (White)

Topic: Representing Sound

Q122. What is the standard sampling rate for CD-quality digital audio?

- A. 22.05 kHz
- B. 44.1 kHz
- C. 48 kHz
- D. 96 kHz

Correct Option: B (44.1 kHz)

Topic: Representing Video

Q123. The frequency at which consecutive images are displayed in a video is called the:

- A. Sampling rate
- B. Frame rate
- C. Bit rate
- D. Refresh rate

Correct Option: B (Frame rate)

Topic: Binary Arithmetic

Q124. What is the binary sum of 101 + 011?

- A. 1000
- B. 1110
- C. 1100
- D. 1011

Correct Option: A (1000)

Topic: 2's Complement Notation

Q125. In an 8-bit 2's complement system, what is the representation of -1?

- A. 10000001
- B. 11111110
- C. 11111111
- D. 00000001

Correct Option: C (11111111)

Topic: Fractions in Binary

Q126. In a binary fraction, the position immediately to the right of the radix point represents the value:

- A. 1/10
- B. 1/2
- C. 1/4
- D. 1/8

Correct Option: B (1/2)

Topic: Floating Point Notation

Q127. The floating-point representation is analogous to which mathematical format?

- A. Scientific notation
- B. Roman numerals
- C. Hexadecimal notation
- D. Prime factorization

Correct Option: A (Scientific notation)

Topic: Data Compression

Q128. Huffman coding is a widely used algorithm for which type of compression?

- A. Lossy compression
- B. Lossless compression
- C. Audio-only compression
- D. Video-only compression

Correct Option: B (Lossless compression)

Topic: CPU Basic

Q129. Which of these is NOT a primary component of a standard Central Processing Unit?

- A. ALU
- B. Control Unit
- C. Registers
- D. Main Memory

Correct Option: D (Main Memory)

Topic: CPU Basic

Q130. The speed at which a CPU executes instructions is typically governed by a:

- A. System Clock
- B. Bus Width
- C. Cache Size
- D. Hard Drive speed

Correct Option: A (System Clock)

Topic: Machine Instruction Categories

Q131. Which instruction type performs arithmetic operations like ADD and SUBTRACT?

- A. Data Transfer
- B. Control
- C. Logic
- D. Arithmetic

Correct Option: D (Arithmetic)

Topic: Program Execution

Q132. During the 'Fetch' phase of the machine cycle, where is the instruction retrieved from?

- A. ALU
- B. Hard Drive
- C. Main Memory
- D. Output Device

Correct Option: C (Main Memory)

Topic: Program Execution

Q133. Which CPU register temporarily holds the instruction currently being decoded and executed?

- A. Program Counter
- B. Instruction Register
- C. Accumulator
- D. Memory Data Register

Correct Option: B (Instruction Register)

Topic: Logic Operators

Q134. A bitwise XOR operation between a bit pattern and a mask of all 1s will result in:

- A. All 1s
- B. All 0s
- C. The inverse of the original pattern
- D. The original pattern unchanged

Correct Option: C (The inverse of the original pattern)

Topic: Rotation and Shift

Q135. Which operation shifts bits to the right and discards the rightmost bit, inserting a 0 on the left?

- A. Logical right shift
- B. Arithmetic left shift
- C. Circular shift
- D. Rotate right

Correct Option: A (Logical right shift)

Topic: Role of Controller

Q136. Which port standard allows daisy-chaining of up to 127 peripheral devices?

- A. VGA
- B. Serial
- C. USB
- D. Parallel

Correct Option: C (USB)

Topic: Direct Memory Access

Q137. Why is DMA advantageous for overall system performance?

- A. It increases clock speed
- B. It bypasses the CPU for memory transfers, freeing the CPU
- C. It expands hard drive capacity
- D. It cools the motherboard

Correct Option: B (It bypasses the CPU for memory transfers, freeing the CPU)

Topic: Operating Systems: History

Q138. An OS interface that relies entirely on text-based commands is known as a:

- A. GUI
- B. CLI
- C. API
- D. IDE

Correct Option: B (CLI)

Topic: Software Classification

Q139. Programs created specifically to assist users in performing everyday, real-world tasks are called:

- A. System software
- B. Application software
- C. Firmware
- D. Drivers

Correct Option: B (Application software)

Topic: Operating Systems: Architecture

Q140. The user interface that utilizes icons, windows, and a pointing device is commonly referred to as a:

- A. CLI
- B. Batch Interface
- C. GUI
- D. Shell

Correct Option: C (GUI)

Topic: Process Administration

Q141. The activity of managing multiple processes by switching the CPU rapidly among them is called:

- A. Paging
- B. Multiprogramming / Multitasking
- C. Booting
- D. Spooling

Correct Option: B (Multiprogramming / Multitasking)

Topic: Process Administration

Q142. In operating system terminology, a program currently in execution is officially termed a:

- A. Thread
- B. Process
- C. Script
- D. Routine

Correct Option: B (Process)

Topic: Operating Systems: Deadlock

Q143. Which of the following is a necessary condition for a deadlock to occur?

- A. Preemption
- B. Mutual Exclusion
- C. Unlimited resources
- D. Single-tasking

Correct Option: B (Mutual Exclusion)

Topic: Networking and the Internet

Q144. A network topology where all computers are connected to a single central cable is called a:

- A. Star topology
- B. Ring topology
- C. Bus topology
- D. Mesh topology

Correct Option: C (Bus topology)

Topic: Networking and the Internet

Q145. Which device is used to connect two distinctly different networks together and direct traffic between them?

- A. Hub
- B. Switch
- C. Router
- D. Repeater

Correct Option: C (Router)

Topic: Algorithm: Formal Definition

Q146. Which of these is NOT a mandatory characteristic of a valid algorithm?

- A. Unambiguous
- B. Infinite execution
- C. Ordered steps
- D. Produces a result

Correct Option: B (Infinite execution)

Topic: Algorithm: Representation

Q147. In pseudocode conventions, what symbol is typically used to denote an assignment operation?

- A. ==
- B. <-
- C. =>
- D. ++

Correct Option: B (<-)

Topic: Algorithm: Discovery

Q148. The problem-solving approach that involves analyzing a complex problem and breaking it down into smaller, manageable subproblems is known as:

- A. Top-down design
- B. Bottom-up design
- C. Object-oriented design
- D. Agile design

Correct Option: A (Top-down design)

Topic: Iterative Structures

Q149. Which control structure executes a block of code repeatedly as long as a specified condition remains true?

- A. If-Else statement
- B. While loop
- C. Switch statement
- D. Sequence

Correct Option: B (While loop)

Topic: Iterative Structures

Q150. A loop structure that inadvertently lacks a proper termination condition and runs forever is called an:

- A. Infinite loop
- B. Nested loop
- C. Pretest loop
- D. Posttest loop

Correct Option: A (Infinite loop)

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Topic: Search Engines

Q151. Which logical operator can be used in search engines to find pages containing at least one of several words?

- A. AND
- B. OR
- C. NOT
- D. XOR

Correct Option: B (OR)

Topic: Search Engines

Q152. What is the purpose of the 'site:' operator in a search engine query?

- A. To search for a specific file type
- B. To search within a specific website or domain
- C. To find related websites
- D. To search for exact phrases

Correct Option: B (To search within a specific website or domain)

Topic: Roots of Computing

Q153. Who invented the calculating machine known as the Pascaline?

- A. Charles Babbage
- B. Blaise Pascal
- C. Alan Turing
- D. John Napier

Correct Option: B (Blaise Pascal)

Topic: Roots of Computing

Q154. Which pioneer developed a punch-card tabulation machine for the 1890 US Census?

- A. Herman Hollerith
- B. Ada Lovelace
- C. John von Neumann
- D. Steve Jobs

Correct Option: A (Herman Hollerith)

Topic: Roots of Computing

Q155. The Atanasoff-Berry Computer (ABC) is significant because it was the first computing device to use:

- A. Transistors
- B. Microprocessors
- C. Vacuum tubes
- D. Mechanical relays

Correct Option: C (Vacuum tubes)

Topic: Bits

Q156. In a binary sequence, the bit positioned at the far right is often referred to as the:

- A. Most Significant Bit (MSB)
- B. Least Significant Bit (LSB)
- C. Sign Bit
- D. Parity Bit

Correct Option: B (Least Significant Bit (LSB))

Topic: Bits

Q157. In a binary sequence, the bit positioned at the far left is known as the:

- A. Most Significant Bit (MSB)
- B. Least Significant Bit (LSB)
- C. Check Bit
- D. End Bit

Correct Option: A (Most Significant Bit (MSB))

Topic: Boolean Operations

Q158. Which Boolean gate produces a 1 output ONLY if both inputs are 0?

- A. NOR
- B. NAND
- C. XOR
- D. OR

Correct Option: A (NOR)

Topic: Boolean Operations

Q159. A truth table for a 3-input Boolean function will have how many rows of input combinations?

- A. 3
- B. 6
- C. 8
- D. 9

Correct Option: C (8)

Topic: Hexadecimal Notation

Q160. What is the hexadecimal representation of the 8-bit binary pattern 1111 0000?

- A. 0F
- B. F0
- C. AA
- D. FF

Correct Option: B (F0)

Topic: Hexadecimal Notation

Q161. The hexadecimal system is a positional numeral system with a base of:

- A. 2
- B. 8
- C. 10
- D. 16

Correct Option: D (16)

Topic: Main Memory

Q162. Which type of memory sits very close to the CPU and holds frequently accessed data to speed up processing?

- A. Virtual Memory
- B. Cache Memory
- C. Secondary Storage
- D. ROM

Correct Option: B (Cache Memory)

Topic: Main Memory

Q163. Memory that loses its data when the power is shut off is strictly described as:

- A. Non-volatile
- B. Static
- C. Volatile
- D. Dynamic

Correct Option: C (Volatile)

Topic: Magnetic Systems

Q164. The formatting process prepares a magnetic disk by dividing its surface into tracks and:

- A. Clusters
- B. Sectors
- C. Platters
- D. Cylinders

Correct Option: B (Sectors)

Topic: Flash Drives

Q165. Solid State Drives (SSDs) primarily utilize which type of non-volatile memory?

- A. SRAM
- B. DRAM
- C. NAND flash
- D. Magnetic tape

Correct Option: C (NAND flash)

Topic: Mass Storage

Q166. Which mass storage medium allows only sequential access to its data?

- A. Hard Disk Drive
- B. Magnetic Tape
- C. CD-ROM
- D. Flash Drive

Correct Option: B (Magnetic Tape)

Topic: Representing Text

Q167. Which Unicode transformation format uses a variable length of 1 to 4 bytes per character?

- A. UTF-8
- B. UTF-16
- C. UTF-32
- D. ASCII

Correct Option: A (UTF-8)

Topic: Representing Images

Q168. The clarity or detail of a bitmapped image, determined by the number of pixels, is referred to as its:

- A. Color depth
- B. Resolution
- C. Amplitude
- D. Frequency

Correct Option: B (Resolution)

Topic: Representing Images

Q169. In the RGB color scheme, what color is produced if Red and Green are at maximum and Blue is zero?

- A. Cyan
- B. Magenta
- C. Yellow
- D. White

Correct Option: C (Yellow)

Topic: Representing Sound

Q170. The height of a sound wave, which determines its volume or loudness, is called its:

- A. Frequency
- B. Pitch
- C. Amplitude
- D. Sampling rate

Correct Option: C (Amplitude)

Topic: Representing Video

Q171. Software or hardware that compresses and decompresses digital video is known as a:

- A. Router
- B. Codec
- C. Compiler
- D. Kernel

Correct Option: B (Codec)

Topic: Binary Arithmetic

Q172. What is the binary subtraction result of 1010 - 0011?

- A. 0101
- B. 0111
- C. 0110
- D. 1001

Correct Option: B (0111)

Topic: 2's Complement Notation

Q173. To find the 2's complement of a binary number, you invert all the bits and then:

- A. Add 1
- B. Subtract 1
- C. Add 0
- D. Multiply by 2

Correct Option: A (Add 1)

Topic: Fractions in Binary

Q174. The binary fraction 0.01 is equivalent to what decimal fraction?

- A. 1/2
- B. 1/4
- C. 1/8
- D. 1/10

Correct Option: B (1/4)

Topic: Floating Point Notation

Q175. In floating-point representation, the part of the number containing the significant digits is the:

- A. Exponent
- B. Radix point
- C. Mantissa
- D. Sign bit

Correct Option: C (Mantissa)

Topic: Data Compression

Q176. Which dictionary-based compression algorithm is heavily used in the GIF and TIFF image formats?

- A. Huffman
- B. LZW
- C. Run-length
- D. MP3

Correct Option: B (LZW)

Topic: Data Compression

Q177. Files with a .zip extension are created using which broad category of compression?

- A. Lossy
- B. Lossless
- C. Differential
- D. Audio

Correct Option: B (Lossless)

Topic: CPU Basic

Q178. Which specialized register in the ALU holds intermediate arithmetic and logic results?

- A. Program Counter
- B. Instruction Register
- C. Accumulator
- D. Memory Address Register

Correct Option: C (Accumulator)

Topic: CPU Basic

Q179. The main printed circuit board that houses the CPU, memory, and connectors is the:

- A. Breadboard
- B. Motherboard
- C. Expansion card
- D. Control Unit

Correct Option: B (Motherboard)

Topic: Machine Instruction Categories

Q180. Instructions that facilitate communication between the CPU and peripheral devices fall into which category?

- A. Arithmetic/Logic
- B. Control
- C. Data Transfer
- D. I/O Instructions

Correct Option: D (I/O Instructions)

Topic: Machine Instruction Categories

Q181. Which instruction explicitly stops the execution of a program?

- A. JUMP
- B. HALT
- C. STORE
- D. LOAD

Correct Option: B (HALT)

Topic: Program Execution

Q182. During the machine cycle, the Decode step is exclusively handled by the:

- A. ALU
- B. Main Memory
- C. Control Unit
- D. System Bus

Correct Option: C (Control Unit)

Topic: Logic Operators

Q183. If you want to ensure a specific bit is set to 1 in a bit pattern, which masking operation should you use?

- A. AND
- B. OR
- C. XOR
- D. NOT

Correct Option: B (OR)

Topic: Role of Controller

Q184. Slots on the motherboard that allow you to plug in extra controllers or cards are called:

- A. Ports
- B. Registers
- C. Expansion slots
- D. Buses

Correct Option: C (Expansion slots)

Topic: Communication media

Q185. Transferring data one bit at a time over a single wire is known as:

- A. Parallel communication
- B. Serial communication
- C. Broadband communication
- D. Multiplexed communication

Correct Option: B (Serial communication)

Topic: Communication media

Q186. The maximum rate at which data can be transferred over a communication channel is its:

- A. Latency
- B. Bandwidth
- C. Seek time
- D. Frequency

Correct Option: B (Bandwidth)

Topic: Operating Systems: Basic Concepts

Q187. An operating system dedicated to a specific task requiring absolute timing reliability is a:

- A. Batch OS
- B. Time-sharing OS
- C. Real-time OS
- D. Desktop OS

Correct Option: C (Real-time OS)

Topic: Operating Systems: Architecture

Q188. The portion of an operating system that directly interacts with the human user is the:

- A. Kernel
- B. Shell / User Interface
- C. Device Driver
- D. Scheduler

Correct Option: B (Shell / User Interface)

Topic: Operating Systems: Architecture

Q189. Small programs that translate OS commands into instructions a specific hardware device can understand are:

- A. Kernels
- B. Utilities
- C. Device Drivers
- D. Compilers

Correct Option: C (Device Drivers)

Topic: Process Administration

Q190. When a process is waiting for an I/O operation to complete, it is typically placed in which state?

- A. Running
- B. Ready
- C. Waiting / Blocked
- D. Terminated

Correct Option: C (Waiting / Blocked)

Topic: Operating Systems: Deadlock

Q191. Which technique allows the OS to intercept output meant for a printer and store it on disk temporarily?

- A. Paging
- B. Spooling
- C. Swapping
- D. Booting

Correct Option: B (Spooling)

Topic: Networking and the Internet

Q192. A network confined to a relatively small geographic area like a single building is a:

- A. WAN
- B. MAN
- C. LAN
- D. PAN

Correct Option: C (LAN)

Topic: Networking and the Internet

Q193. A standardized set of rules governing the exchange of data between network devices is called a:

- A. Topology
- B. Packet
- C. Protocol
- D. Router

Correct Option: C (Protocol)

Topic: Algorithm: Representation

Q194. The standard execution structure where instructions are performed one after another in order is:

- A. Sequence
- B. Selection
- C. Iteration
- D. Recursion

Correct Option: A (Sequence)

Topic: Iterative Structures

Q195. A loop structure where the termination test is evaluated after the body has executed at least once is a:

- A. Pretest loop
- B. Posttest loop
- C. Infinite loop
- D. Nested loop

Correct Option: B (Posttest loop)

Topic: Data Structures

Q196. In a one-dimensional array, the integer used to identify a specific element's position is its:

- A. Value
- B. Pointer
- C. Index
- D. Node

Correct Option: C (Index)

Topic: Sorting Algorithms

Q197. Which algorithm sorts by repeatedly stepping through the list, comparing adjacent elements, and swapping them if they are in the wrong order?

- A. Insertion Sort
- B. Bubble Sort
- C. Binary Sort
- D. Sequential Sort

Correct Option: B (Bubble Sort)

Topic: Algorithm Efficiency

Q198. In the worst-case scenario, how many comparisons might a sequential search make on a list of N items?

- A. $\log_2(N)$
- B. $N/2$
- C. N
- D. N^2

Correct Option: C (N)

Topic: Algorithm Efficiency

Q199. The notation used in computer science to express the upper bound or worst-case complexity of an algorithm is:

- A. Omega notation
- B. Theta notation
- C. Big O notation
- D. Alpha notation

Correct Option: C (Big O notation)

Topic: Operating Systems: Architecture

Q200. The component of the OS responsible for managing files, directories, and storage space allocation is the:

- A. Process Manager
- B. Memory Manager
- C. File Manager
- D. Device Manager

Correct Option: C (File Manager)

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Topic: Introduction to Computer Science

Q201. Which field focuses specifically on creating machines capable of mimicking human intelligence?

- A. Robotics
- B. Artificial Intelligence
- C. Data Science
- D. Software Engineering

Correct Option: B (Artificial Intelligence)

Topic: Introduction to Computer Science

Q202. What term is generally used to describe the physical, tangible components of a computer?

- A. Software
- B. Firmware
- C. Hardware
- D. Shareware

Correct Option: C (Hardware)

Topic: Search Engines

Q203. Which search query uses an asterisk (*) to represent unknown or missing words in a phrase?

- A. Exact match search
- B. Wildcard search
- C. Boolean search
- D. Domain search

Correct Option: B (Wildcard search)

Topic: Search Engines

Q204. Which symbol is used in search engines to search for a specific price or currency?

- A. @
- B. #
- C. \$
- D. %

Correct Option: C (\$)

Topic: Roots of Computing

Q205. Who is historically credited with writing the first algorithm intended to be processed by a machine?

- A. Alan Turing
- B. Charles Babbage
- C. Ada Lovelace
- D. Grace Hopper

Correct Option: C (Ada Lovelace)

Topic: Roots of Computing

Q206. What does the acronym ENIAC stand for?

- A. Electronic Numerical Integrator and Computer
- B. Electrical Node Interface and Controller
- C. Electronic Network Identifier and Calculator
- D. Electrical Number Interpreter and Compiler

Correct Option: A (Electronic Numerical Integrator and Computer)

Topic: Bits

Q207. How many bits are contained in a single nibble?

- A. 2
- B. 4
- C. 8
- D. 16

Correct Option: B (4)

Topic: Bits

Q208. Which number system uses only the digits 0 and 1?

- A. Decimal
- B. Octal
- C. Hexadecimal
- D. Binary

Correct Option: D (Binary)

Topic: Boolean Operations

Q209. The output of an XOR gate is 0 only if the two inputs are:

- A. Both 1
- B. Both 0
- C. Different
- D. Identical (both 0 or both 1)

Correct Option: D (Identical (both 0 or both 1))

Topic: Boolean Operations

Q210. Which Boolean operation is considered the logical inverse or negation?

- A. AND
- B. OR
- C. XOR
- D. NOT

Correct Option: D (NOT)

Topic: Hexadecimal Notation

Q211. In the hexadecimal system, the letter 'C' represents which decimal number?

- A. 10
- B. 11
- C. 12
- D. 13

Correct Option: C (12)

Topic: Hexadecimal Notation

Q212. What is the highest decimal value represented by a single hexadecimal digit?

- A. 9
- B. 15
- C. 16
- D. 31

Correct Option: B (15)

Topic: Storing a Bit

Q213. The process of periodically recharging a capacitor in dynamic memory to prevent data loss is called:

- A. Swapping
- B. Paging
- C. Refreshing
- D. Formatting

Correct Option: C (Refreshing)

Topic: Main Memory

Q214. ROM is generally used to store which essential system program?

- A. Word Processor
- B. BIOS / Bootstrap loader
- C. Web Browser
- D. File Manager

Correct Option: B (BIOS / Bootstrap loader)

Topic: Main Memory

Q215. Which unit is typically used to measure the capacity of a modern computer's main memory (RAM)?

- A. Kilobytes (KB)
- B. Megabytes (MB)
- C. Gigabytes (GB)
- D. Bits

Correct Option: C (Gigabytes (GB))

Topic: Magnetic Systems

Q216. The concentric rings on the surface of a magnetic disk platter are called:

- A. Sectors
- B. Cylinders
- C. Tracks
- D. Clusters

Correct Option: C (Tracks)

Topic: Magnetic Systems

Q217. The delay in waiting for the desired sector of a disk to rotate under the read/write head is called:

- A. Seek time
- B. Transfer rate
- C. Rotational delay (Latency)
- D. Access time

Correct Option: C (Rotational delay (Latency))

Topic: Optical System

Q218. Which technology typically uses a red laser to read and write data?

- A. CD
- B. DVD
- C. Blu-ray
- D. Floppy Disk

Correct Option: B (DVD)

Topic: Flash Drives

Q219. USB flash drives primarily use which type of non-volatile memory?

- A. DRAM
- B. SRAM
- C. NAND flash memory
- D. ROM

Correct Option: C (NAND flash memory)

Topic: Mass Storage

Q220. Which sequential mass storage medium is commonly used for large-scale enterprise data archiving?

- A. Floppy Disk
- B. CD-ROM
- C. Magnetic tape
- D. Flash Drive

Correct Option: C (Magnetic tape)

Topic: Representing Text

Q221. How many bits were used by the original, basic ASCII encoding scheme?

- A. 7
- B. 8
- C. 16
- D. 32

Correct Option: A (7)

Topic: Representing Text

Q222. Extended ASCII uses how many bits to include additional special characters?

- A. 7
- B. 8
- C. 16
- D. 24

Correct Option: B (8)

Topic: Representing Images

Q223. The smallest addressable element or unit of a digital image display is a:

- A. Vector
- B. Node
- C. Pixel
- D. Dot

Correct Option: C (Pixel)

Topic: Representing Images

Q224. Which color model is the standard for representing colors on computer displays and TVs?

- A. CMYK
- B. RGB
- C. Grayscale
- D. HSV

Correct Option: B (RGB)

Topic: Representing Sound

Q225. What standard protocol is widely used to encode instructions for digital musical instruments?

- A. MP3
- B. WAV
- C. MIDI
- D. FLAC

Correct Option: C (MIDI)

Topic: Representing Video

Q226. The number of individual still images displayed per second in a digital video is called the:

- A. Resolution
- B. Bit rate
- C. Frame rate
- D. Refresh rate

Correct Option: C (Frame rate)

Topic: Binary Arithmetic

Q227. What is the result of adding 010 and 010 in binary?

- A. 011
- B. 100
- C. 101
- D. 110

Correct Option: B (100)

Topic: 2's Complement Notation

Q228. What is the 2's complement of the binary pattern 00000000?

- A. 11111111
- B. 10000000
- C. 00000000
- D. 00000001

Correct Option: C (00000000)

Topic: Fractions in Binary

Q229. The binary fraction 0.100 is mathematically equal to what decimal value?

- A. 0.125
- B. 0.25
- C. 0.5
- D. 0.75

Correct Option: C (0.5)

Topic: Floating Point Notation

Q230. In standard IEEE 754 floating point format, a sign bit of '0' indicates that the number is:

- A. Negative
- B. Zero
- C. Positive
- D. Infinity

Correct Option: C (Positive)

Topic: Data Compression

Q231. MP3 is a highly popular, lossy compression standard specifically designed for what type of data?

- A. Video
- B. Audio
- C. Images
- D. Text

Correct Option: B (Audio)

Topic: Data Compression

Q232. Which type of compression method is fully reversible, ensuring no original data is lost?

- A. Lossy
- B. Lossless
- C. Destructive
- D. Heuristic

Correct Option: B (Lossless)

Topic: CPU Basic

Q233. The internal pathway that carries the memory address from the CPU to RAM is called the:

- A. Data Bus
- B. Control Bus
- C. Address Bus
- D. System Bus

Correct Option: C (Address Bus)

Topic: CPU Basic

Q234. The size of a CPU's general-purpose registers generally defines its:

- A. Clock speed
- B. Word size
- C. Cache capacity
- D. Core count

Correct Option: B (Word size)

Topic: Machine Instruction Categories

Q235. An instruction designed to simply move data from one register to another is classified as a:

- A. Logic instruction
- B. Control instruction
- C. Data Transfer instruction
- D. Arithmetic instruction

Correct Option: C (Data Transfer instruction)

Topic: Machine Instruction Categories

Q236. A conditional jump instruction changes the flow of the program based on:

- A. The system clock
- B. The result of a previous operation (flags)
- C. The size of the hard drive
- D. The user's input

Correct Option: B (The result of a previous operation (flags))

Topic: Program Execution

Q237. Which special-purpose CPU register holds the exact instruction currently being decoded?

- A. Program Counter
- B. Accumulator
- C. Instruction Register
- D. Memory Data Register

Correct Option: C (Instruction Register)

Topic: Logic Operators

Q238. Applying an AND mask consisting entirely of 0s to any bit pattern will result in:

- A. The original pattern
- B. The inverted pattern
- C. All 1s
- D. All 0s

Correct Option: D (All 0s)

Topic: Role of Controller

Q239. The standard, versatile interface that largely replaced older serial and parallel ports on modern PCs is:

- A. VGA
- B. PS/2
- C. USB
- D. SCSI

Correct Option: C (USB)

Topic: Communication media

Q240. Which physical transmission medium uses pulses of light to transmit data at high speeds?

- A. Coaxial cable
- B. Twisted pair
- C. Fiber optic cable
- D. Microwave

Correct Option: C (Fiber optic cable)

Topic: Communication media

Q241. A data transmission mode that allows data to flow in both directions simultaneously is called:

- A. Simplex
- B. Half-duplex
- C. Full-duplex
- D. Multiplex

Correct Option: C (Full-duplex)

Topic: Operating Systems: History

Q242. Early computer systems that executed one job completely before starting the next utilized:

- A. Multitasking
- B. Batch processing
- C. Time-sharing
- D. Real-time processing

Correct Option: B (Batch processing)

Topic: Software Classification

Q243. Proprietary software available initially for free but typically requiring payment for full functionality is called:

- A. Freeware
- B. Open-source
- C. Shareware
- D. Public domain

Correct Option: C (Shareware)

Topic: Operating Systems: Architecture

Q244. The core component of an OS that directly manages hardware and system resources is the:

- A. Shell
- B. Kernel
- C. Compiler
- D. Interpreter

Correct Option: B (Kernel)

Topic: Process Administration

Q245. A process that has finished execution but still has a lingering entry in the process table is termed a:

- A. Orphan process
- B. Zombie process
- C. Blocked process
- D. Daemon

Correct Option: B (Zombie process)

Topic: Operating Systems: Deadlock

Q246. Resolving a deadlock by forcefully halting a process and taking back its resources is known as:

- A. Preemption
- B. Mutual exclusion
- C. Spooling
- D. Paging

Correct Option: A (Preemption)

Topic: Networking and the Internet

Q247. The unique numerical identifier assigned to every device connected to a network is its:

- A. MAC address
- B. IP address
- C. URL
- D. Port number

Correct Option: B (IP address)

Topic: Algorithm: Representation

Q248. In standard algorithm flowcharting, a parallelogram shape is strictly used to represent:

- A. Processing steps
- B. Decisions
- C. Input/Output operations
- D. Start/Stop points

Correct Option: C (Input/Output operations)

Topic: Iterative Structures

Q249. A programming loop that is completely contained within the body of another loop is called a:

- A. Infinite loop
- B. Recursive loop
- C. Nested loop
- D. Sequential loop

Correct Option: C (Nested loop)

Topic: Algorithm Efficiency

Q250. An algorithm whose execution time grows proportionally in a straight line with the input size has a Big O complexity of:

- A. $O(1)$
- B. $O(\log N)$
- C. $O(N)$
- D. $O(N^2)$

Correct Option: C ($O(N)$)

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Topic: Data Storage: Error Handling

Q251. Which of the following is a simple method used to detect errors in transmitted data by adding an extra bit?

- A. Check sum
- B. Parity bit
- C. Hamming code
- D. Encryption bit

Correct Option: B (Parity bit)

Topic: Data Storage: Error Handling

Q252. In an odd parity system, if the data bits are 1011, the parity bit added will be:

- A. 0
- B. 1
- C. Cannot be determined
- D. None

Correct Option: A (0)

Topic: Data Storage: Mass Storage

Q253. What does RAID stand for in the context of data storage?

- A. Redundant Array of Independent Disks
- B. Random Access Interactive Data
- C. Read/write Access In Drives
- D. Rapid Action Information Deployment

Correct Option: A (Redundant Array of Independent Disks)

Topic: Roots of Computing

Q254. Which organization developed the first commercial computer in the United States, the UNIVAC I?

- A. IBM
- B. Eckert-Mauchly Computer Corporation
- C. Apple
- D. Microsoft

Correct Option: B (Eckert-Mauchly Computer Corporation)

Topic: Roots of Computing

Q255. Moore's Law originally stated that the number of transistors on a microchip doubles approximately every:

- A. 6 months
- B. 12 to 24 months
- C. 5 years
- D. 10 years

Correct Option: B (12 to 24 months)

Topic: Main Memory

Q256. Which of the following cache memory levels is typically the smallest but the fastest, being built directly into the CPU core?

- A. L1 Cache
- B. L2 Cache
- C. L3 Cache
- D. RAM

Correct Option: A (L1 Cache)

Topic: Representing Images

Q257. Which lossless image format is widely used on the web and supports transparent backgrounds?

- A. JPEG
- B. BMP
- C. PNG
- D. TIFF

Correct Option: C (PNG)

Topic: Representing Sound

Q258. What is the typical bit depth for standard CD-quality audio?

- A. 8-bit
- B. 16-bit
- C. 24-bit
- D. 32-bit

Correct Option: B (16-bit)

Topic: Representing Video

Q259. Which compression standard is commonly associated with high-definition video formats like Blu-ray?

- A. MPEG-1
- B. MPEG-2
- C. MPEG-4 / H.264
- D. GIF

Correct Option: C (MPEG-4 / H.264)

Topic: Binary Arithmetic

Q260. If you add 1 to the highest possible positive number in a fixed-size 2's complement system, what happens?

- A. Underflow
- B. Overflow resulting in a negative number
- C. The system halts
- D. The number stays the same

Correct Option: B (Overflow resulting in a negative number)

Topic: Fractions in Binary

Q261. What is the decimal equivalent of the binary fraction 0.11?

- A. 0.25
- B. 0.50
- C. 0.75
- D. 1.5

Correct Option: C (0.75)

Topic: Data Compression

Q262. Run-length encoding is most effective for compressing which type of data?

- A. Highly detailed photographs
- B. Text with few repeating characters
- C. Images with large blocks of solid color
- D. Complex audio files

Correct Option: C (Images with large blocks of solid color)

Topic: CPU Architecture Philosophies

Q263. A microprocessor design that utilizes a large, complex set of instructions is known as:

- A. RISC
- B. CISC
- C. ARM
- D. MIPS

Correct Option: B (CISC)

Topic: Machine Cycle

Q264. The speed of the machine cycle is often measured in:

- A. Hertz (Hz)
- B. Bytes
- C. Pixels
- D. Baud

Correct Option: A (Hertz (Hz))

Topic: Program Execution

Q265. Which internal CPU component performs the actual mathematical and logical operations during the 'Execute' phase?

- A. Control Unit
- B. Registers
- C. ALU
- D. Cache

Correct Option: C (ALU)

Topic: Logic Operators

Q266. In bit manipulation, an OR operation with a mask of 1s is commonly used to:

- A. Clear bits to 0
- B. Set bits to 1
- C. Invert bits
- D. Check bit status

Correct Option: B (Set bits to 1)

Topic: Direct Memory Access

Q267. Which of the following components coordinates DMA transfers between peripherals and main memory?

- A. DMA Controller
- B. ALU
- C. System Clock
- D. Cache Manager

Correct Option: A (DMA Controller)

Topic: Communication media

Q268. Which of these is an example of an unguided (wireless) communication medium?

- A. Coaxial cable
- B. Fiber optics
- C. Microwave transmission
- D. Twisted pair

Correct Option: C (Microwave transmission)

Topic: Operating Systems: History

Q269. In early batch processing systems, what medium was primarily used to submit jobs?

- A. Magnetic Tape
- B. Punched Cards
- C. Floppy Disks
- D. Optical Discs

Correct Option: B (Punched Cards)

Topic: Operating Systems: Architecture

Q270. The part of the OS that resides in memory at all times and provides essential basic services is the:

- A. Utility
- B. Shell
- C. Kernel
- D. Application

Correct Option: C (Kernel)

Topic: Process Administration

Q271. A situation where a process is perpetually denied necessary resources to proceed is called:

- A. Deadlock
- B. Starvation
- C. Paging
- D. Spooling

Correct Option: B (Starvation)

Topic: Process Administration

Q272. The specific OS module that decides which ready process will execute next is the:

- A. Memory Manager
- B. Device Driver
- C. Scheduler / Dispatcher
- D. Compiler

Correct Option: C (Scheduler / Dispatcher)

Topic: Memory Management

Q273. Which memory management technique divides memory into fixed-size blocks called frames?

- A. Segmentation
- B. Paging
- C. Fragmentation
- D. Swapping

Correct Option: B (Paging)

Topic: Operating Systems: Deadlock

Q274. Which of the following is NOT one of the Coffman conditions required for a deadlock?

- A. Mutual exclusion
- B. Hold and wait
- C. No preemption
- D. Multithreading

Correct Option: D (Multithreading)

Topic: Networking and the Internet

Q275. Which network topology connects every node to every other node, providing high redundancy?

- A. Star
- B. Bus
- C. Ring
- D. Mesh

Correct Option: D (Mesh)

Topic: Networking and the Internet

Q276. What operates at the physical layer to regenerate and amplify network signals over long distances?

- A. Router
- B. Repeater
- C. Gateway
- D. Firewall

Correct Option: B (Repeater)

Topic: Networking and the Internet

Q277. Which protocol is responsible for resolving human-readable domain names into IP addresses?

- A. HTTP
- B. FTP
- C. DNS
- D. SMTP

Correct Option: C (DNS)

Topic: Networking and the Internet

Q278. An IPv4 address consists of how many bits?

- A. 16
- B. 32
- C. 64
- D. 128

Correct Option: B (32)

Topic: Networking and the Internet

Q279. To accommodate the growing number of devices, IPv6 was introduced. How many bits does an IPv6 address use?

- A. 32
- B. 64
- C. 128
- D. 256

Correct Option: C (128)

Topic: Networking and the Internet

Q280. Which software application is used to access and display webpages from the World Wide Web?

- A. Web Server
- B. Search Engine
- C. Web Browser
- D. Operating System

Correct Option: C (Web Browser)

Topic: Algorithm: Formal Definition

Q281. An algorithm must always finish after a finite number of steps. This characteristic is known as:

- A. Unambiguity
- B. Effectiveness
- C. Termination / Finiteness
- D. Precision

Correct Option: C (Termination / Finiteness)

Topic: Algorithm: Representation

Q282. In a flowchart, what shape is used to denote the start or end of the process?

- A. Rectangle
- B. Diamond
- C. Parallelogram
- D. Oval / Pill shape

Correct Option: D (Oval / Pill shape)

Topic: Iterative Structures

Q283. In structured programming, which construct allows the selection of one out of many possible execution paths based on a specific value?

- A. While loop
- B. Switch / Case statement
- C. Sequence
- D. For loop

Correct Option: B (Switch / Case statement)

Topic: Recursive Structure

Q284. A crucial requirement for a recursive function is that it must have a:

- A. Global variable
- B. Base case to terminate
- C. For loop
- D. Pointer

Correct Option: B (Base case to terminate)

Topic: Sorting Algorithms

Q285. Which sorting algorithm works by finding the smallest element from the unsorted portion and swapping it with the first unsorted element?

- A. Insertion Sort
- B. Selection Sort
- C. Bubble Sort
- D. Merge Sort

Correct Option: B (Selection Sort)

Topic: Searching Algorithms

Q286. A mandatory precondition for using the Binary Search algorithm is that the list must be:

- A. Small in size
- B. Containing integers only
- C. Sorted
- D. Unordered

Correct Option: C (Sorted)

Topic: Algorithm Efficiency

Q287. Which of the following time complexities represents the fastest, most efficient execution regardless of input size?

- A. $O(N)$
- B. $O(1)$
- C. $O(\log N)$
- D. $O(N^2)$

Correct Option: B ($O(1)$)

Topic: Algorithm Efficiency

Q288. The time complexity of a standard Binary Search algorithm is:

- A. $O(1)$
- B. $O(N)$
- C. $O(\log N)$
- D. $O(N^2)$

Correct Option: C ($O(\log N)$)

Topic: Data Structures

Q289. A data structure where elements follow a First-In-First-Out (FIFO) principle is called a:

- A. Stack
- B. Queue
- C. Tree
- D. Graph

Correct Option: B (Queue)

Topic: Data Structures

Q290. A data structure where elements are added and removed from the same end (Last-In-First-Out, LIFO) is a:

- A. Queue
- B. Array
- C. Linked List
- D. Stack

Correct Option: D (Stack)

Topic: Web Page Development

Q291. Which markup language is the standard for structuring content on the World Wide Web?

- A. Python
- B. Java
- C. HTML
- D. C++

Correct Option: C (HTML)

Topic: Web Page Development

Q292. In HTML, which tag is typically used to create a hyperlink?

A.

B.

Correct Option: B ()

Topic: Web Page Development

Q293. What technology is primarily used alongside HTML to style and format the visual layout of a webpage?

A. CSS

B. SQL

C. PHP

D. XML

Correct Option: A (CSS)

Topic: Networking and the Internet

Q294. A system that acts as an intermediary for requests from clients seeking resources from other servers is a:

A. Proxy server

B. Database server

C. File server

D. Mail server

Correct Option: A (Proxy server)

Topic: Software Classification

Q295. Software whose source code is freely available for anyone to inspect, modify, and enhance is known as:

A. Freeware

B. Shareware

C. Proprietary software

D. Open-source software

Correct Option: D (Open-source software)

Topic: Security

Q296. Malicious software designed to block access to a computer system until a sum of money is paid is called:

A. Adware

B. Ransomware

C. Spyware

D. Worm

Correct Option: B (Ransomware)

Topic: Security

Q297. A security system designed to monitor and control incoming and outgoing network traffic based on predetermined security rules is a:

A. Switch

B. Router

C. Firewall

D. Repeater

Correct Option: C (Firewall)

Topic: Data Manipulation

Q298. An operation that reverses the truth value of a single bit is also known as a:

A. Shift

B. Mask

C. Complement / Flip

D. Rotation

Correct Option: C (Complement / Flip)

Topic: Program Execution

Q299. Which of these CPU architectural designs increases instruction throughput by processing multiple instructions at different stages of the machine cycle simultaneously?

- A. Virtualization
- B. Multiprogramming
- C. Pipelining
- D. Swapping

Correct Option: C (Pipelining)

Topic: Operating Systems: Basic Concepts

Q300. The process of starting up a computer and loading the operating system into main memory is called:

- A. Formatting
- B. Booting
- C. Executing
- D. Compiling

Correct Option: B (Booting)